Introductory Mathematics

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Introductory Mathematics (A07559)

Short Title:	Introductory Mathematics
Faculty	Science and Computing
Department	Science
Cluster	Mathematics and Physics
Author	ADALYWALSH
Credits	5
Level:	Introductory

Description of Module / Aims

This module will introduce the student to linear and non-linear algebra. It will explore the basics of trigonometry and how it relates to waves. There is an emphasis throughout this module on the role of mathematics in the sciences, including real world applications.

Programmes

MTHS-0001	BSc (Common Entry) (WD_SSCIE_D)	1 / 1 / M
MTHS-0001	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	1 / 1 / M
MTHS-0001	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	1 / 1 / M
MTHS-0001	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	1 / 1 / M
MTHS-0001	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	1 / 1 / M
MTHS-0001	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	1 / 1 / M
MTHS-0001	BSc (Hons) in Science (Common Entry) (WD_SSCCM_B)	1 / 1 / M
MTHS-0001	BSc in Food Science with Business (WD_SFDSC_D)	1 / 1 / M
MTHS-0001	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	1 / 1 / M
MTHS-0001	BSc in Pharmaceutical Science (WD_SPHSC_D)	1 / 1 / M

Indicative Content

- Arithmetic: review of arithmetic operations; fractions; significant figures; decimal places; percentages and ratios
- Introduction to algebra: simplifying algebraic expressions; factorisation and algebraic fractions
- Transposition and solution of equations: rearranging a formula; solving linear equations; solving simultaneous equations; solving quadratic equations
- Introduction to functions: straight line function; polynomial function; rational function; exponential function; logarithmic function
- Angles: measuring angles; length of an arc and area of a sector
- Introduction to trigonometry: trigonometric ratios; finding the angle given one of the ratios; extended definition of trigonometric ratios; trigonometric functions and their graphs
- Trigonometric identities and equations: solution of triangles using Pythagoras' theorem, sine and cosine rule; area of a triangle
- Application of trigonometry to waves: general equation of a sine wave; construction of a sine wave from a circle

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Use and manipulate routine algebraic and trigonometric calculations.
2. Construct linear, non-linear and trigonometric equations of varying complexity.
3. Sketch linear, polynomial, rational, growth and trigonometric functions.
4. Interpret linear, polynomial, rational, growth and trigonometric functions in a meaningful way.
5. Construct mathematical expressions for problems of an applied nature in the science environment.
6. Explain results obtained from problems of an applied nature in the science environment.

Learning and Teaching Methods

- Lectures.
- Online exercises.
- Technology to enhance the learning experience.
- Self-directed learning is emphasised.
- Students are supported with notes, online material and feedback.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	4,5,6
Continuous Assessment	30%	
In-Class Assessment	10%	1,2,3
Tutorial/Problem Sheets	20%	1,2,3

Assessment Criteria

<40%: Inability to identify and describe key concepts of algebra and trigonometry.
40%-49%: Ability to identify and describe key concepts of algebra and trigonometry.
50%-59%: Ability to discuss key concepts of algebra and trigonometry clearly and interpret their relative importance.
60%-69%: Ability to apply solutions to problems based on algebra and trigonometry in a range of relevant contexts. An ability to employ a comprehensive range of specialised skills.
70%-100%: All the above to an excellent level, with the ability to analyse and design solutions to algebraic and trigonometric problems to a high standard, for a range of complex or unseen problems.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Online Delivery	12	
Independent Learning	87	

Essential Material

- Croft, A. and R. Davison. _Foundation Maths_. 5th Edition. England: Pearson Education Limited, 2010.

Supplementary Material

- "Loughborough University, U.K." MathCenter. www.mathcenter.ac.uk
- Aufmann, R.N., V.C. Barker and R.D. Nation. _College Algebra and Trigonometry_. 8th Edition. Boston: Houghton Mifflin, 2001.
- Bird, J.O. and A.J.C May. _Technician Mathematics 1_. 3rd Edition. Malaysia: Longman, 1994.
- Bird, J.O. and A.J.C. May. _Technician Mathematics 2_. 3rd Edition. Malaysia: Longman, 1994.
- Bird, J.O. and A.J.C. May. _Technician Mathematics 3_. 2nd Edition. Malaysia: Longman , 1994.

Requested Resources

- Room Type: Computer Lab